

CDFD Runtime Paper 06: CLI-First Platform and Future Web Architecture

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Abstract

Executive synthesis. The platform direction is CLI first, web app second. The CLI is for serious engine work: reproducible simulations, paper diagnostics, debugging, automation, and saved results. The future web app is a visual layer over the same backend, not a separate physics system. This paper defines that boundary and explains how experiment outputs flow into future visual interfaces.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Ordering Principle

The platform order is:

1. core engine;

2. CLI;
3. API server;
4. web app.

The current repository now has the first two surfaces in active form. The web app under `webapp/` remains useful, but it calls the shared runtime backend rather than duplicate logic.

3 CLI Responsibilities

The CLI supports:

- listing registered domains;
- running domain demos;
- validating CDFL models;
- executing CDFL models;
- exporting JSON result envelopes;
- preserving finite-value and provenance metadata.

This is the credibility layer. It gives papers exact commands instead of vague claims that "the engine was run."

4 Future CDFD Studio

The future web interface is CDFD Studio: a visual environment for model building, simulation running, graph viewing, field visualization, coherence and dissipation monitoring, result browsing, and export.

The first public demos are non-medical and visual:

- heat diffusion and thermal flow;
- traffic or infrastructure routing;
- resource-flow networks;
- predator-prey/ecology models;
- photonic or adaptive-surface demos.

Medical applications require stricter validation, privacy, and user-safety boundaries and are not the first public web surface.

5 API Boundary

The API exposes command-like operations:

- validate a CDFL model;
- run a CDFL model;
- retrieve saved result envelopes;
- inspect finite audits;
- list experiments and selected outputs;
- export reports.

Each endpoint calls `runtime/runner.py` or a later service wrapper around it.

6 API Keys and the AI Boundary

Application keys belong at the application boundary, not inside the physics kernel. The runtime can check whether a calling app presents an approved key, record that key as a redacted fingerprint in provenance, and then run CDFL or domain commands. It does not call an LLM, store LLM credentials, or generate application-facing responses. That layer belongs in CDFD Studio, clinical apps, research notebooks, or other governed products built above the engine.

This split keeps the engine clean: `cdfd.py` and `runtime/runner.py` own deterministic execution, finite audits, and result envelopes; applications own authentication, user identity, LLM routing, rate limits, privacy, and workflow policy.

7 Vacuum OS

Vacuum OS (VOS) is the proposed operating system layer above CDFD Runtime. VOS is not a second engine. It is the orchestration shell for projects that need apps, CDFL packages, user spaces, experiment queues, provenance stores, and controlled AI assistance. In this architecture, CDFD Runtime remains the mathematical kernel; VOS becomes the operating environment that launches models, manages keys, schedules runs, stores outputs, and connects approved applications to the same CDFD command backend.

8 Experiment Visualization

The experiment outputs already define future panels:

- vacuum hysteresis: M_s residual heatmap and recovery curve;
- knot stability: stability-index trace toward the χ^* attractor;
- OOL phase: $\Lambda = 1$ phase boundary;
- frontier sweep: pattern distribution and candidate triage table;

- gigamarathon sweep: domain coverage dashboard.

The web app presents these as model diagnostics and falsification targets, not as empirical proof.

9 Conclusion

The platform makes one thing unmistakable: CDFD is executable. The CLI establishes that discipline. The web app later makes the same discipline visible, interactive, and understandable without weakening the runtime boundary.

References

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